T.C.
BAHÇEŞEHİR UNIVERSITY

AIN4312 Special Topics in Artificial Intelligence Engineering II 25/26 (1)

Explainable AI (XAI) System for Autonomous Vehicles

Student Name	Student Number	Department
Mohamed Ahmed **Allabani**	**2105990**	**Artificial Intelligence Engineering**
Mohamed Aiman Mohamed **Alkhozendar**	**2283149**	**Artificial Intelligence Engineering**
Masa Yousef **Bokhary**	**2287036**	**Artificial Intelligence Engineering**
Abdullah Hani Abdellatif **Al-Shobaki**	**2284612**	**Artificial Intelligence Engineering**
Abdullah Shaher **Salamoun**	**2267205**	**Artificial Intelligence Engineering**

ISTANBUL, 5 January 2026

STUDENT DECLARATION

By submitting this report, as partial fulfillment of the requirements of the AIN 4312 course, the students promise on penalty of failure of the course that

- they have given credit to and declared (by citation), any work that is not their own (e.g. parts of the report that is copied/pasted from the Internet, design or construction performed by another person, etc.);
- they have not received unpermitted aid for the project design, construction, report or presentation;
- they have not falsely assigned credit for work to another student in the group, and not take credit for work done by another student in the group.

ABSTRACT

Autonomous driving systems increasingly rely on complex artificial intelligence models whose internal decision-making processes are often difficult to interpret. This lack of transparency creates challenges for validation, debugging, safety assessment, and user trust. In this project, an Explainable Artificial Intelligence (XAI)–based autonomous vehicle perception and decision-support system is designed and evaluated using a simulation-driven approach.

The system is implemented within the CARLA autonomous driving simulator and integrates RGB camera and LiDAR sensors to enable robust environment perception. Camera-based vehicle detection is performed using the YOLOv5 deep learning model, while LiDAR-based information complements visual perception to improve detection reliability. To address interpretability challenges, Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP) are incorporated to provide visual and feature-level explanations for model predictions and driving-related decisions.

The system architecture follows a modular design that separates simulation, perception, explainability, and visualization components, enabling systematic development and evaluation. Experimental results show that the proposed system accurately detects surrounding vehicles in diverse simulated driving scenarios while generating meaningful explanations for AI-driven outputs. The integration of explainability mechanisms enhances transparency and supports greater trust and reliability in autonomous vehicle systems.

Key Words: Explainable Artificial Intelligence (XAI), Autonomous Vehicles, CARLA Simulator, YOLOv5, LiDAR, Grad-CAM, SHAP, Computer Vision

TABLE OF CONTENTS

ABSTRACT iii
TABLE OF CONTENTS iv
LIST OF TABLES v
LIST OF FIGURES v
LIST OF ABBREVIATIONS vi
1. OVERVIEW 1
1.1. Identification of the need 1
1.2. Definition of the problem 1
1.3. Conceptual solutions 2
1.4. Physical architecture 3
2. WORK PLAN 5
2.1. Work Breakdown Structure (WBS) 5
2.2. Responsibility Matrix (RM) 5
2.3. Project Network (PN) 7
2.4. Gantt chart 7
2.6. Risk assessment 8
3. SUB-SYSTEMS 10
3.1. Simulation and Sensor Setup Sub-system 10
3.2. AI Perception and Explainability Sub-system 11
4. INTERGRATION AND EVALUATION 13
4.1. Integration 13
4.2. AI Perception and Explainability Sub-system 13
5. SUMMARY AND CONCLUSION 14
ACKNOWLEDGEMENTS 15
REFERENCES 16
APPENDIX A 17
APPENDIX B 18

LIST OF TABLES

Table 1. Comparison of the three conceptual solutions 3
Table 2. Responsibility Matrix for the team 6
Table 3. Gantt chart for the implementation phase of the project 8
Table 4. Risk matrix 9
Table 5. Risk assessment 9

LIST OF FIGURES

Figure 1. Interface diagram for the system 4
Figure 2. Process chart for the system 4
Figure 3. Work breakdown structure for the project 5
Figure 4. The project network 7

LIST OF ABBREVIATIONS

AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
CARLA	Car Learning to Act Autonomous Driving Simulator
CNN	Convolutional Neural Network
YOLO	You Only Look Once
YOLOv5	You Only Look Once version 5
LiDAR	Light Detection and Ranging
Gram-CAM	Gradient-weighted Class Activation Mapping
SHAP	SHapley Additive exPlanations
API	Application Programming Interface
WBS	Work Breakdown Structure
RM	Responsibility Matrix
PN	Project Network

1. OVERVIEW

Artificial Intelligence (AI) has become a core component of modern autonomous vehicle systems, enabling perception, decision-making, and control in complex driving environments. Advances in computer vision and deep learning have significantly improved the ability of autonomous systems to detect objects, interpret scenes, and respond in real time. However, these systems often rely on black box models whose internal reasoning processes are difficult to interpret.

This project addresses this challenge by focusing on Explainable Artificial Intelligence (XAI) within the context of autonomous driving. By integrating explainability techniques into perception and decision-making modules, the project aims to enhance system transparency, support validation and debugging, and improve trust in AI-driven autonomous vehicle behavior.

1.1. Identification of the need

As AI becomes increasingly integrated into autonomous vehicle systems, the demand for transparency and accountability has grown. Explainability is essential to ensure safety, public trust, and regulatory compliance. When AI models make critical driving decisions, such as emergency braking or obstacle avoidance, understanding the reasoning behind these actions is necessary for validation and ethical assurance.

Without explainability, developers and evaluators lack insight into why specific decisions were made, which increases safety risks and complicates system verification. Therefore, there is a clear need for autonomous driving systems that provide interpretable and transparent decision-making mechanisms.

1.2. Definition of the problem

Despite the high performance of modern AI-based perception systems, their lack of interpretability presents significant limitations for safety-critical applications such as autonomous driving. When AI models make decisions related to braking, steering, or obstacle avoidance, it is often unclear which input features or environmental factors influenced those decisions. This opacity complicates debugging, performance evaluation, and regulatory approval.

The core problem addressed in this project is the absence of integrated explainability mechanisms in autonomous vehicle perception systems. Without clear explanations, developers and evaluators cannot reliably assess whether system decisions are reasonable, robust, or biased. Therefore, there is a need for an autonomous driving framework that combines accurate perception with interpretable and transparent AI decision-making.

1.2.1. Functional requirements

The functional requirements define the core capabilities that the proposed autonomous driving system must provide in order to achieve its intended objectives. These requirements describe the essential system behaviors related to perception, explainability, and decision support within the autonomous driving framework.

Based on the identified problem and project objectives, the following functional requirements were defined to guide system design and implementation:

- Develop an AI model capable of performing key autonomous driving functions such as obstacle detection and lane following.
- Integrate explainability mechanisms to visually and textually describe AI decision-making.
- Provide real-time visual feedback highlighting which inputs influenced each driving action.
- Store decision logs with explanation metadata for later analysis and verification.

1.2.2. Performance requirements

The performance requirements specify the quantitative criteria that the proposed system must satisfy to ensure reliable and efficient operation in real-time autonomous driving scenarios. These requirements focus on system responsiveness, accuracy, and stability when combining perception, decision-making, and explainability components.

Since explainability mechanisms introduce additional computational overhead, the system must maintain acceptable performance levels while executing both prediction and explanation processes concurrently. Based on these considerations, the following performance requirements were defined:

- Maintain real-time decision latency under 100 ms per frame.
- Achieve at least 90% detection accuracy on driving scene components.
- Preserve system stability while running both prediction and explanation modules simultaneously.
- Demonstrate less than 10% performance degradation when XAI features are active.

1.2.3. Constraints

The development of the proposed system is subject to technical, economic, and operational constraints. From a technical perspective, the system must operate within the computational limitations of available hardware, as real-time perception and explainability techniques can be resource-intensive. Additionally, reliance on simulation-based environments limits the realism of sensor data compared to real-world deployments.

From an economic and social perspective, the project is constrained by limited development time and resources, requiring efficient use of open-source tools and existing frameworks. Environmentally, the use of simulation reduces the need for physical testing, lowering energy consumption and material waste. Socially, the system must promote ethical AI usage by ensuring that explanations are understandable and do not mislead users or stakeholders.

1.3. Conceptual solutions

The proposed system integrates computer vision and explainable AI algorithms to interpret model outputs. The conceptual solution includes the following components:

- Use of a fine-tuned YOLOv5 model for visual scene understanding.
- Implementation of Grad-CAM for image-based explanations and SHAP for feature attribution.

- Connection of perception outputs to a control decision module responsible for lane keeping, braking, and steering actions.
- Development of a visual dashboard to display real-time explanations within the simulator.

This conceptual solution combines AI-based perception with explainability mechanisms to ensure both performance and transparency. By integrating computer vision models with XAI techniques, the system is designed to perform autonomous driving functions while providing human-interpretable justifications for its decisions.

1.3.1. Literature Review

Recent research highlights the growing importance of Explainable Artificial Intelligence in autonomous systems. Selvaraju et al. (2017) introduced Grad-CAM, which visualizes image regions that most influence model predictions. Lundberg and Lee (2017) proposed SHAP, a method based on game theory for estimating feature importance. Studies such as Kim et al. (2020) demonstrate that integrating explainability tools with perception models improves user trust and facilitates debugging. This project builds upon these works by combining Grad-CAM and SHAP within a unified explainability framework for autonomous driving.

1.3.2. Concepts

Multiple conceptual solutions were evaluated to address the challenge of integrating explainable artificial intelligence into autonomous vehicle systems. These concepts differ in terms of system complexity, computational cost, explainability depth, and suitability for real-time operation.

The evaluated concepts include a perception-only autonomous system without explainability, an explainability-enhanced perception system using post-hoc visualization methods, and a fully integrated perception and explainability framework. These concepts were compared based on cost, complexity, performance, and explainability capability, as summarized in Table 1. Based on this comparison, Concept 3 was selected as it provides strong explainability capabilities while maintaining acceptable performance and system complexity.

Table 1. Comparison of the three conceptual solutions.

	Concept 1	Concept 2	Concept 3
Description	Autonomous perception without explainability	Perception with post-hoc explainability	Integrated perception & explainability framework
Cost	Low	medium	medium
Complexity	Low	medium	High
Performance	High	medium	High
Explainability	None	Partial	high

1.4. Physical architecture

The physical architecture describes the high-level organization of the system components and the interfaces between them. Although the proposed autonomous vehicle system is entirely software-based, a physical architecture representation is required to illustrate how major modules interact and exchange data within the system.

The architecture is designed in a modular manner, separating simulation, perception, explainability, and visualization components. This separation enables independent development, testing, and maintenance of each module while ensuring seamless integration through well-defined interfaces.

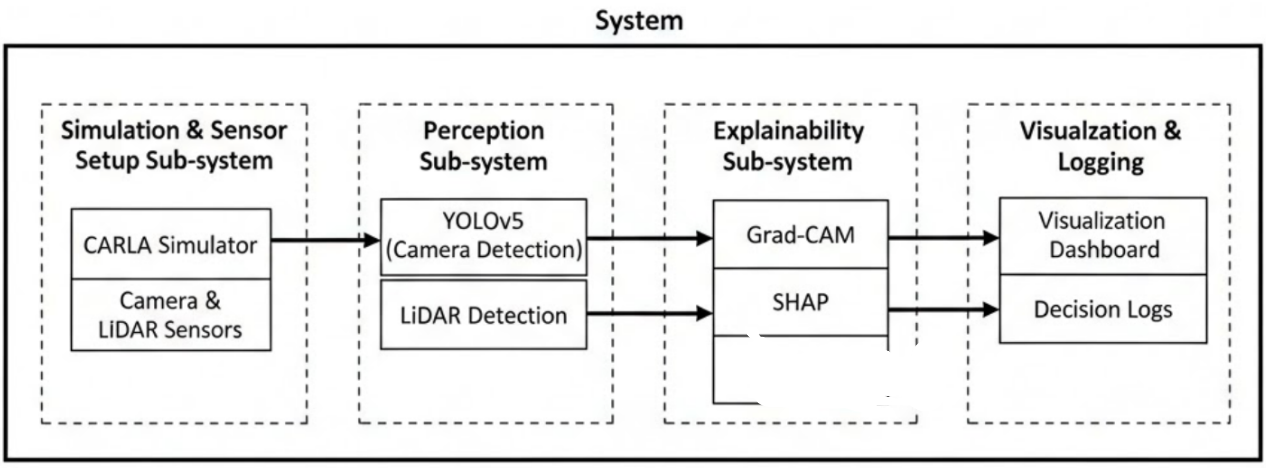


Figure 1. Interface diagram for the system.

The interface diagram illustrates the flow of sensor data and control information between the main system components. Sensor data generated within the simulation environment is transmitted to the perception modules, where camera and LiDAR inputs are processed in parallel. The resulting perception outputs are then forwarded to explainability modules and visualization components.

The overall process flow of the system is summarized in the process chart shown in Figure 2. This chart presents the sequential stages of simulation, perception, explainability, decision support, and visualization, highlighting how data moves through the system from input acquisition to interpretable output generation.

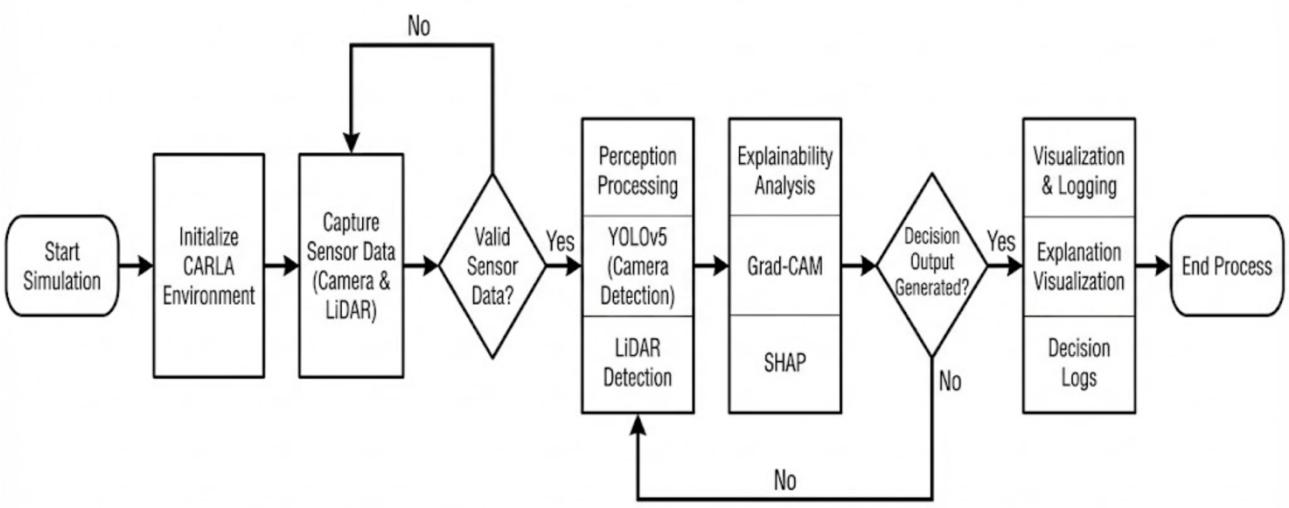


Figure 2. Process chart for the system

2. WORK PLAN

This section describes the project planning methodology followed during the development of the Explainable AI system for autonomous vehicles. The work plan defines how project activities were structured, scheduled, and coordinated throughout the semester to ensure systematic progress and timely completion of both technical implementation and documentation tasks.

The planning approach emphasizes task decomposition, responsibility assignment, scheduling, and risk management. Key planning artifacts including the Work Breakdown Structure (WBS), Responsibility Matrix, Project Network, Gantt chart, and risk assessment are used to provide a comprehensive overview of project execution.

2.1. Work Breakdown Structure (WBS)

A structured planning approach was adopted to manage the multidisciplinary nature of the project, which includes simulation setup, perception system development, explainability implementation, system integration, and evaluation. The Work Breakdown Structure was developed to decompose the project into manageable tasks and sub-tasks, enabling systematic tracking and coordination.

The WBS illustrates the hierarchical organization of project activities and clarifies the relationships between major development phases. As illustrated in Figure 3, the Work Breakdown Structure provides a visual representation of task decomposition from high-level objectives down to individual implementation activities.

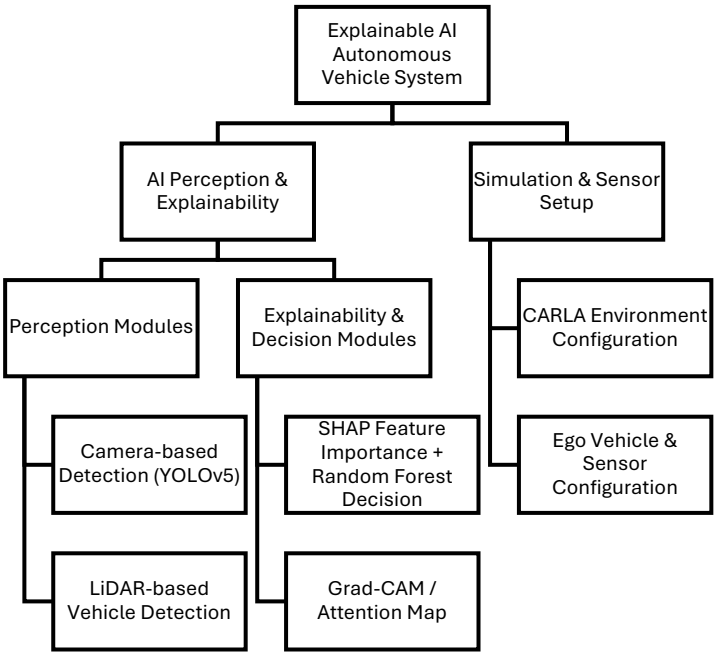


Figure 3. Work breakdown structure for the project.

2.2. Responsibility Matrix (RM)

The Responsibility Matrix defines the allocation of tasks among team members involved in the project. Its purpose is to clearly identify responsibility and support roles for each major activity, ensuring accountability and effective collaboration throughout the project lifecycle.

Responsibilities are distributed based on each team member's contribution to system development, including simulation setup, perception module implementation, explainability mechanisms, system integration, and documentation. As shown in Table 2, the notation **R** indicates the team member primarily responsible for a task, while **S** denotes a supporting role. This structured allocation helps avoid task overlap and supports parallel development.

Table 2. Responsibility Matrix for the team

Task	Abdullah Al-Shobaki	Mohammed Alkozendar	Abdullah Salamoun	Mohammed Allabani	Masa Bokhary
System architecture and planning	**R**	S	S	S	S
CARLA simulation setup	**R**	S	S		
Sensor setup and configuration (Camera and LiDAR)	S	**R**			
Perception module development (YOLOv5 + basic tracking)	S	**R**			
Explainability integration (Grad-CAM + SHAP)		**R**	S		
Visualization dashboard and explanation overlays		S	**R**		
Performance testing and speed vs interpretability analysis				**R**	
Evaluation scenarios and results analysis	**R**	S	S	**R**	
Report writing and formatting	**R**	S	S	S	**R**
Version control and final presentation materials	S	S			**R**

R = Responsible, S = Support

2.3. Project Network (PN)

The Project Network illustrates the logical sequence and dependencies between major project activities. It is used to represent the order in which tasks must be executed and to identify prerequisite relationships that influence scheduling and coordination.

As shown in Figure 4, the project network reflects both sequential and parallel relationships among development phases such as simulation and sensor setup, perception module implementation, explainability integration, system integration, and evaluation. This representation helps ensure that dependent tasks begin only after prerequisite activities are completed.

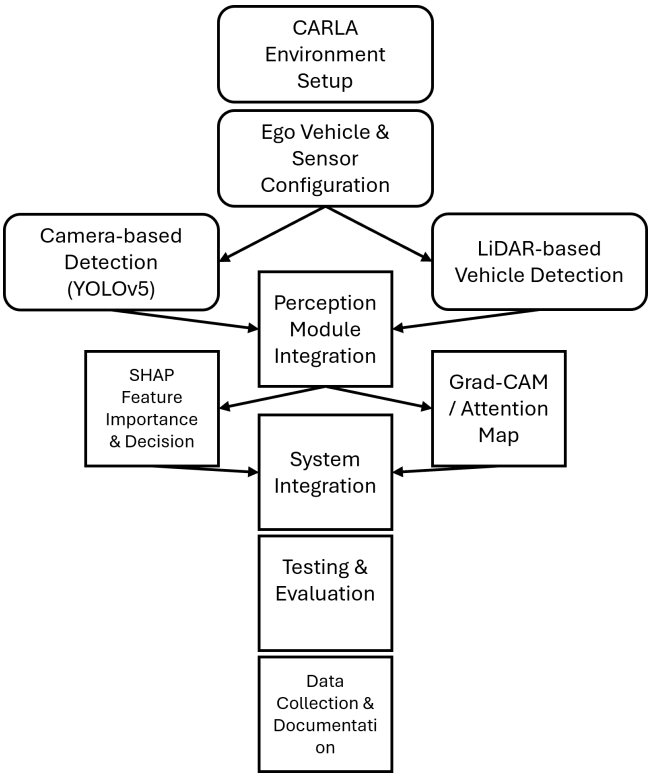


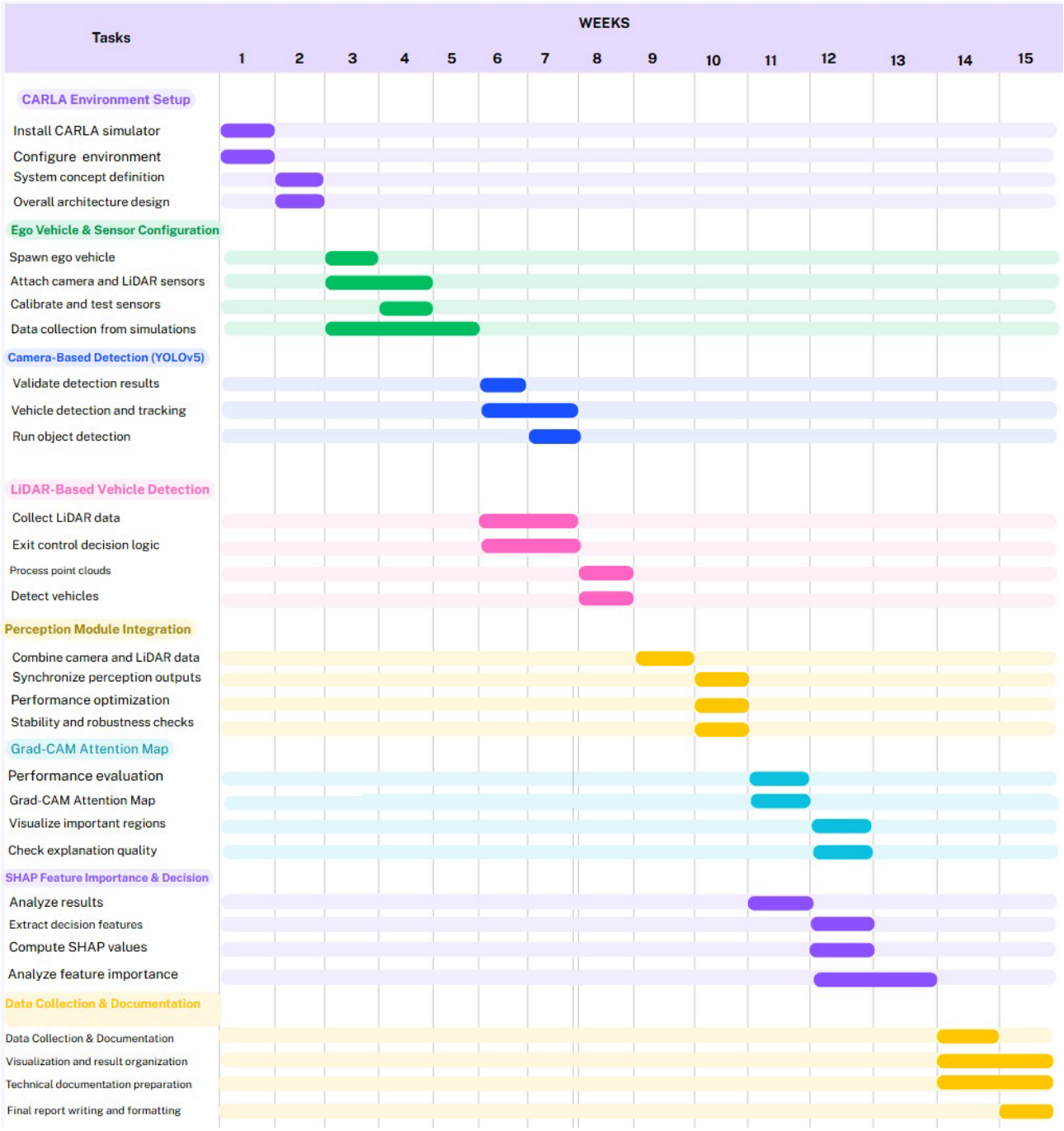
Figure 4. The project network.

2.4. Gantt chart

The Gantt chart presents the project schedule over the 15-week semester and illustrates the timing, duration, and overlap of the main development activities. It reflects the logical sequencing of tasks derived from the Work Breakdown Structure and Project Network while highlighting opportunities for parallel development.

Task responsibilities shown in the Gantt chart are assigned in accordance with the Responsibility Matrix presented in Section 2.2. As summarized in Table 3, the Gantt chart provides a clear timeline for implementation, evaluation, and documentation activities, supporting effective time management throughout the project.

Table 3. Gantt chart for the implementation phase of the project.



2.6. Risk assessment

Risk assessment was conducted to identify potential technical, operational, and project management risks that could affect the successful completion of the Explainable AI autonomous vehicle system. Identified risks were evaluated based on their probability of occurrence and the severity of their potential impact on project outcomes.

A qualitative risk matrix was used to categorize risks into different risk levels, enabling prioritization and proactive mitigation planning. Table 4 presents the risk matrix used for

assessment, while Table 5 summarizes the identified risks along with their probability, severity, risk level, and mitigation strategies.

Table 4. Risk matrix

RISK LEVEL		Severity of the event on the project success		
		Minor	Moderate	Major
Probability of the event occuring	Unlikely	VERY LOW	LOW	MEDIUM
	Possible	LOW	MEDIUM	HIGH
	Likely	MEDIUM	HIGH	VERY HIGH

VERY LOW	This event is very low risk and so does not require any plan for mitigation. In the unlikely event that it does occur there will be only a minor effect on the project.
LOW	This event is low-risk; a preliminary study on a plan of action to recover from the event can be performed and noted.
MEDIUM	This event presents a signficant risk; a plan of action to recover from it should be made and resources sourced in advance.
HIGH	This event presents a very signficant risk. Consider changing the product design/project plan to reduce the risk; else a plan of action for recovery should be made and resources sourced in advance.
VERY HIGH	**This is an unacceptable risk. The product design/project plan must be changed to reduce the risk to an acceptable level.**

Table 5. Risk assessment

Risk	Risk Level	Mitigation Strategy	Contingency Plan
CARLA simulation instability	High	Use stable CARLA versions, incremental scenario testing, and modular development.	Switch to simplified scenarios or previous stable configurations to continue development.
YOLOv5 detection performance degradation	High	Fine-tune model parameters and validate detections under multiple traffic conditions.	Rely more heavily on LiDAR-based detection for safety-critical decisions.
High computational load reducing real-time performance	High	Optimize code, reduce input resolution, and limit SHAP update frequency.	Disable real-time explainability temporarily and log explanations offline.
LiDAR and camera integration issues	Medium	Perform independent sensor testing and staged integration.	Fall back to single-sensor operation for debugging and validation.
Limited testing time before submission	Medium	Prioritize critical test scenarios and automate evaluation where possible.	Reduce non-essential features and focus on core system functionality.

3. SUB-SYSTEMS

This section describes the main sub-systems that constitute the Explainable AI autonomous vehicle system developed in this project. The overall system is divided into multiple sub-systems based on functionality and development responsibility, enabling modular design, independent implementation, and systematic evaluation.

The project is organized into two primary sub-systems: the **Simulation and Sensor Setup Sub-system** and the **AI Perception and Explainability Sub-system**. Each sub-system is described in terms of its requirements, technologies and methods, conceptual design, physical architecture, materialization, and evaluation.

3.1. Simulation and Sensor Setup Sub-system

The Simulation and Sensor Setup Sub-system is responsible for creating the virtual driving environment and configuring the ego vehicle and its sensors. This sub-system provides the foundation upon which perception, decision-making, and explainability modules operate.

By isolating simulation and sensor configuration from higher-level intelligence modules, this sub-system enables controlled experimentation, repeatability, and safe testing of autonomous driving scenarios without the risks associated with real-world deployment.

3.1.1. Requirements

The primary requirements of this sub-system include the ability to simulate realistic urban driving scenarios, configure an ego vehicle with multiple sensors, and generate reliable sensor data for downstream processing. The simulation environment must support dynamic traffic participants and repeatable experiments.

Additionally, the sub-system must allow flexible configuration of camera and LiDAR sensors and ensure synchronization between sensor outputs to support accurate perception and analysis by higher-level sub-systems.

3.1.2. Technologies and methods

This sub-system was implemented using the CARLA autonomous driving simulator, which provides high-fidelity urban environments, realistic vehicle dynamics, and programmable traffic behavior. CARLA was used to configure the ego vehicle, define sensor placements, and simulate various driving scenarios.

RGB camera and LiDAR sensors were attached to the ego vehicle using CARLA's sensor application programming interface (API). Sensor data was streamed in real time to perception modules, enabling continuous evaluation of detection and explainability algorithms.

3.1.3. Conceptualization

Conceptually, this sub-system functions as the data generation layer of the overall system. It abstracts the complexity of real-world data collection by providing controlled, reproducible simulation scenarios with configurable sensor inputs.

The separation of simulation and perception allows independent development and testing of each component, improving system modularity, maintainability, and experimental reliability.

3.1.4. Physical architecture

The physical architecture consists of a CARLA server responsible for running the simulation environment and a client module that controls the ego vehicle and receives sensor data. This server–client architecture supports scalable simulation and efficient data streaming.

Sensor data is transmitted to processing modules running on the same computational platform, ensuring low-latency communication and synchronized data delivery.

3.1.5. Materialization

This sub-system was materialized by configuring multiple CARLA simulation maps, defining traffic scenarios, and attaching RGB camera and LiDAR sensors to the ego vehicle. Sensor parameters such as resolution, field of view, range, and update rate were tuned to balance realism and computational efficiency.

The implemented setup enabled consistent generation of synchronized sensor data across multiple test scenarios used throughout system development and evaluation.

3.1.6. Evaluation

The Simulation and Sensor Setup Sub-system was evaluated based on its stability, realism, and ability to provide consistent and synchronized sensor data. Successful execution of multiple simulation runs without system crashes served as a key evaluation criterion.

The evaluation confirmed that the sub-system met all functional requirements and provided a reliable and stable foundation for perception, decision-making, and explainability components.

3.2. AI Perception and Explainability Sub-system

The AI Perception and Explainability Sub-system is responsible for detecting surrounding vehicles from sensor data, integrating perception outputs, and generating interpretable explanations for AI-driven decisions. This sub-system represents the core intelligence of the autonomous vehicle system.

By combining perception and explainability within a single sub-system, the system ensures that autonomous driving decisions are not only accurate but also transparent and interpretable.

3.2.1. Requirements

The primary requirements of this sub-system include accurate vehicle detection from camera and LiDAR data, robust integration of multi-sensor perception outputs, and the ability to explain model decisions in a transparent and interpretable manner.

The sub-system must support real-time or near real-time processing and generate explanations that are meaningful to developers and system evaluators without significantly degrading system performance.

3.2.2. Technologies and methods

Camera-based vehicle detection was implemented using the YOLOv5 convolutional neural network–based object detection model, which provides efficient and accurate detection from image data. LiDAR-based vehicle detection was implemented using point cloud processing and clustering techniques.

Explainability was achieved using Gradient-weighted Class Activation Mapping (Grad-CAM) for visual attention analysis and SHapley Additive exPlanations (SHAP) for feature importance analysis. These techniques provide complementary visual and numerical insights into model behavior.

3.2.3. Conceptualization

Conceptually, this sub-system integrates perception and explainability to ensure that autonomous vehicle decisions are both reliable and interpretable. Camera and LiDAR detections are processed in parallel and subsequently fused to improve robustness.

Explainability modules operate alongside perception and decision components, enabling continuous interpretation of model outputs without interrupting system functionality.

3.2.4. Physical architecture

The physical architecture consists of multiple processing modules running on a computational platform capable of handling deep learning workloads. Camera images and LiDAR point clouds are processed by dedicated perception modules whose outputs are forwarded to an integration layer.

Explainability modules access intermediate and final model outputs to generate visual and feature-level explanations, which are logged and displayed for analysis and evaluation.

3.2.5. Materialization

This sub-system was materialized by training and integrating the YOLOv5 model for camera-based detection, implementing LiDAR-based detection pipelines, and developing explainability modules using Grad-CAM and SHAP.

An integration layer was implemented to combine perception outputs and support system-level decision-making, enabling seamless interaction between detection, fusion, and explainability components.

3.2.6. Evaluation

Evaluation of the AI Perception and Explainability Sub-system focused on detection accuracy, system robustness, and the interpretability of generated explanations. Detection performance was assessed across multiple simulated driving scenarios.

Explainability outputs were evaluated qualitatively based on clarity, consistency, and relevance to system decisions. The evaluation confirmed that the sub-system successfully detects vehicles and provides meaningful explanations, thereby enhancing transparency and trust in the autonomous vehicle system.

4. INTERGRATION AND EVALUATION

This section describes the integration process of the developed sub-systems and the evaluation methodology used to assess the performance and reliability of the Explainable AI autonomous vehicle system. Integration focuses on ensuring seamless communication between simulation, perception, explainability, and visualization components.

The evaluation process examines both system-level functionality and qualitative performance aspects, including detection accuracy, system stability, and the effectiveness of explainability outputs. Together, integration and evaluation confirm that the complete system operates as intended within the simulation environment.

4.1. Integration

System integration involved combining the Simulation and Sensor Setup Sub-system with the AI Perception and Explainability Sub-system to form a unified autonomous driving framework. Sensor data generated in the simulation environment was streamed in real time to the perception modules, enabling continuous processing and decision support.

The integration process also ensured that explainability modules operated alongside perception and decision components without disrupting real-time execution. Outputs from perception and explainability modules were synchronized and forwarded to a centralized visualization dashboard, allowing real-time monitoring and analysis of system behavior.

4.2. AI Perception and Explainability Sub-system

The integrated system was evaluated using a series of simulated driving scenarios designed to test perception accuracy, system robustness, and explainability quality. Scenarios included varying traffic densities and vehicle interactions to assess system behavior under different conditions.

Evaluation results indicate that the integrated system successfully detects surrounding vehicles and generates consistent and interpretable explanations for AI-driven outputs. The evaluation confirms that the integration approach supports reliable system operation while enhancing transparency and trust in autonomous vehicle decision-making.

5. SUMMARY AND CONCLUSION

In summary, this project presented the design, implementation, and evaluation of an Explainable Artificial Intelligence (XAI) system for autonomous vehicles using a simulation-based approach. The system integrates RGB camera and LiDAR sensors within the CARLA simulator to enable robust vehicle perception while incorporating explainability techniques to address the transparency challenges of modern AI-driven autonomous systems.

The project demonstrated how visual and feature-based explanation methods, specifically Grad-CAM and SHAP, can be embedded into an autonomous driving framework without significantly compromising system performance. Through modular system design and structured integration, the proposed solution supports effective debugging, validation, and interpretation of AI-driven decisions in simulated driving environments.

In conclusion, the results confirm that integrating explainability mechanisms into autonomous vehicle perception and decision-support systems enhances system transparency and interpretability. The developed system successfully detects surrounding vehicles and provides meaningful explanations that clarify how sensor inputs influence AI outputs, thereby supporting greater trust in autonomous driving technologies.

While the project was limited to a simulation environment, it establishes a solid foundation for future research and development of explainable autonomous systems. Future work may extend the proposed framework to real-world datasets, additional driving behaviors, and more advanced explainability techniques, further strengthening the role of XAI in safety-critical autonomous applications.

ACKNOWLEDGEMENTS

We wish to thank our adviser **Prof. Murat Göksin** for his guidance, support, and valuable feedback throughout the development of this project. His suggestions and expertise were instrumental in shaping the technical approach and improving the overall quality of the work.

We would also like to acknowledge the support and cooperation of our team members, as well as the assistance and encouragement provided by friends and peers during the implementation, testing, and documentation phases of the project.

REFERENCES

1. R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, Venice, Italy, pp. 618–626, Oct. 2017.
2. S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems (NeurIPS)*, Long Beach, CA, USA, pp. 4765–4774, 2017.
3. B. Kim, R. Khanna, and O. Koyejo, "Examples are not enough, learn to criticize! Criticism for interpretability," *Advances in Neural Information Processing Systems (NeurIPS)*, Montréal, Canada, pp. 2280–2288, 2016.
4. B. Kim, J. Wattenberg, J. Gilmer, C. Cai, J. Wexler, F. Viégas, and R. Sayres, "Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV)," *Proceedings of the 35th International Conference on Machine Learning (ICML)*, Stockholm, Sweden, pp. 2668–2677, 2018.
5. A. Bojarski et al., "End to end learning for self-driving cars," *arXiv preprint arXiv:1604.07316*, 2016.
6. A. Dosovitskiy et al., "CARLA: An open urban driving simulator," *Proceedings of the 1st Annual Conference on Robot Learning (CoRL)*, Mountain View, CA, USA, pp. 1–16, 2017.
7. J. Redmon and A. Farhadi, "YOLOv3: An incremental improvement," *arXiv preprint arXiv:1804.02767*, 2018.
8. G. Jocher et al., "YOLOv5," *GitHub repository*, 2020. [Online]. Available: https://github.com/ultralytics/yolov5. [Accessed: Jan. 2026].
9. M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you?: Explaining the predictions of any classifier," *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, San Francisco, CA, USA, pp. 1135–1144, 2016.
10. F. Rosique, P. J. Navarro, C. Fernández, and A. Padilla, "A systematic review of perception system and simulators for autonomous vehicles research," *Sensors*, vol. 19, no. 3, pp. 648–670, Feb. 2019.
11. A. B. Arrieta et al., "Explainable Artificial Intelligence (XAI): Concepts, taxonomies, opportunities and challenges toward responsible AI," *Information Fusion*, vol. 58, pp. 82–115, June 2020.
12. A. Omeiza, H. Webb, M. Jirotka, and L. Kunze, "Explanations in autonomous driving: A survey," *IEEE Transactions on Intelligent Transportation Systems*, vol. 23, no. 8, pp. 10142–10162, Aug. 2022.

APPENDIX A

This appendix provides additional implementation details for the Explainable Artificial Intelligence autonomous vehicle system developed in this project. Due to the length and complexity of the software implementation, the complete source code is not included in the main body of the report.

The full implementation, including simulation setup, sensor configuration, perception modules, and explainability components, is provided in the accompanying Jupyter Notebook file submitted with this report. The notebook contains executable code for the CARLA simulation environment, YOLOv5-based vehicle detection, LiDAR processing, and explainability techniques using Grad-CAM and SHAP.

Selected code excerpts were used during development to demonstrate sensor data handling, perception processing, and explanation generation. All code is written in Python and was executed in a simulation-based environment for evaluation and testing purposes.

APPENDIX B

No additional material is included in this appendix.